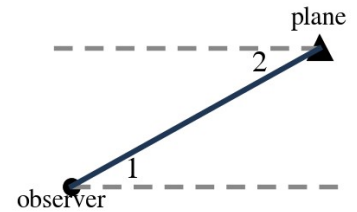


Speak Like A Mathematician:

angle of elevation:

angle of depression:

Why are the two angles in the figure equal?



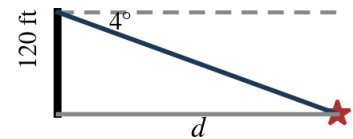
Setting Up Elevation / Depression Problems

1. Draw the right triangle: the _____ distance and the height are the legs; the line of sight is the hypotenuse.
2. Put the angle at the OBSERVER, measured from the _____ line.
3. Almost every problem is a _____ equation: the height and the horizontal distance are opposite and adjacent.

Example 1 - A Ranger Spots a Fire

A ranger 120 ft up a tower sees a fire at an angle of depression of 4° (figure at the right).

- a) Which leg is known, and which is wanted?
- b) Find d to the nearest foot.



Example 2 - Height from a Shadow

A tower casts a 58-m shadow when the angle of elevation from the shadow's tip to the top is 68° (figure at the right).

- a) Find the height to the nearest meter.

Example 3 - A Plane Overhead

- a) A plane flies at 4200 ft. The angle of elevation from the airport to the plane is 31° . Find the horizontal distance to the nearest foot.

Example 4 - Looking Down a Cliff

- a) From the edge of a 96-ft-wide gorge, the angle of depression to the base of the far wall is 47° . Find the depth to the nearest foot.

Example 5 - Two Boats from a Lighthouse

An observer 75 ft above the water sees two boats in line, at angles of depression 51° (near) and 24° (far). Figure at the right.

- a) Find each boat's distance from the lighthouse.
- b) How far apart are the boats?

